

Section 6 Computational Supplement

PARI/GP programs, exact certificates, and GRH status
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1. Purpose and logical status

These files reproduce the computations in Section 6 of the current manuscript. The endpoint programs search in the relevant descent field for principal ideals I satisfying $I \cdot \text{conj}(I) = 2^f$ and then carry out the support test required by the element partition method. The separate certificate PDF records explicit integral solutions of the quadratic, biquadratic, and quartic norm-form equations used to compare the ideal and element endpoints.

The distinction between conditional and unconditional output is essential. The large-field endpoint searches use `bnfinit` without `bnfcertify`; their first-PIP-endpoint values and the resulting nonexistence statements are therefore conditional on GRH. In contrast, the quartic benchmark program calls `bnfcertify` and stops if certification fails, so every benchmark value reported by a completed run is unconditional.

The Diophantine equalities in `section6_diophantine_solutions.pdf` are exact and unconditional. The two exceptional generator checks also verify their final norm identities by exact arithmetic. Nevertheless, equality of the two endpoints remains conditional on GRH because the assertion that the reported exponent is the first PIP endpoint comes from an uncertified large-field class-group computation.

2. File inventory

File	Family	Role
<code>cyclic_type3mod8_1mod8.txt</code>	Case (III)	Fast norm-pruned search when the primes above 2 form a cyclic orbit. Conditional on GRH.
<code>type_3mod8_1mod8.txt</code>	Case (III)	General direct search without the cyclic-orbit assumption. Conditional on GRH.
<code>type_3mod8_5mod8.txt</code>	Case (I)	Large-field PIP and support search in the field L . Conditional on GRH.
<code>type_7mod8_1mod8.txt</code>	Case (II)	Search for $p_1 \equiv 7 \pmod{8}$, $p_2 \equiv 1 \pmod{8}$. Conditional on GRH.
<code>type_7mod8_5mod8.txt</code>	Feng, Theorem 4.1, Case (2)	Search for $p_1 \equiv 7 \pmod{8}$, $p_2 \equiv 5 \pmod{8}$. This family is treated in Section 6 but is not one of the three cases of the current endpoint-extension theorem. Conditional on GRH.
<code>determine_m.txt</code>	Case (III) benchmark	Determines the earlier quartic PIP threshold. Requires <code>bnfcertify = 1</code> ; output is unconditional.
<code>check_11_61.txt</code>	Exceptional Case (I)	Runs the endpoint search for (11,61), recovers generators, and checks $\alpha \cdot \text{conj}(\alpha) = 2^{13}$ exactly.
<code>check_67_97.txt</code>	Exceptional Case (III)	Runs the endpoint search for (67,97), recovers generators, and checks $\alpha \cdot \text{conj}(\alpha) = 2^{37}$ exactly.
<code>section6_diophantine_solutions.pdf</code>	All nonexceptional examples	Typeset exact certificates for the nine remaining parameter triples.

3. Parameters in the computational proposition

The first coordinate below is the reported odd endpoint exponent r . To reproduce a row, edit only the input values near the beginning of the indicated search program.

Family	Triples (r, p_1, p_2)	Endpoint program	Element certificate
Case (I): 3 mod 8, 5 mod 8	(3,19,13); (11,67,13); (13,11,61)	type_3mod8_5mod8.txt	PDF certificate for the first two; check_11_61.txt for the last.
Feng, Theorem 4.1, Case (2): 7 mod 8, 5 mod 8	(7,71,61); (5,79,37); (5,79,61)	type_7mod8_5mod8.txt	PDF certificate.
Current Theorem 3, Case (II): 7 mod 8, 1 mod 8	(3,31,17); (7,71,41)	type_7mod8_1mod8.txt	PDF certificate.
Case (III): 3 mod 8, 1 mod 8	(23,19,97); (7,11,41); (37,67,97)	cyclic_type3mod8_1mod8.txt; use the general program if the cyclic test fails.	PDF certificate for the first two; check_67_97.txt for the last.

4. Running the programs

Use a current PARI/GP installation. From GP, load a file with the read command. For example:

```
\r type_3mod8_5mod8.txt
```

Each search file has a clearly marked input block containing p_1 , p_2 , the starting odd exponent, and a maximum exponent. Some programs also expose e_1 and e_2 . For the Section 6 examples, set $e_1 = e_2 = 1$. The current manuscript explains why conductor $p_1 p_2$ suffices when the primes are not Wieferich primes.

For Case (III), first try `cyclic_type3mod8_1mod8.txt`. It is much faster because it uses the cyclic orbit of primes above 2 together with quartic norm pruning. If the program reports that no transitive automorphism was found, use `type_3mod8_1mod8.txt`, which does not require that orbit assumption.

The large-field programs print the first odd exponent with principal ideal solutions, verify that the recovered full solutions are principal, and report whether the same-support check passes. A completed run should be interpreted with the GRH warning printed at the top of the source file.

5. Benchmark computations

The older Feng and Feng-Liu bounds use a smaller quadratic or quartic field. The quadratic thresholds are obtained from explicit Diophantine equations. For the Case (III) quartic fields, run `determine_m.txt` after entering the desired p_1 and p_2 . The program builds the quartic field, calls `bnfcertify`, and refuses to continue if certification fails. Thus a reported quartic threshold is not GRH-dependent.

6. Exact comparison of the two endpoints

For nine triples, `section6_diophantine_solutions.pdf` gives an explicit integral solution of the relevant norm-form equation and substitutes it into both sides. This provides an unconditional witness x with $x \cdot \text{conj}(x) = 2^r$.

For $(r, p_1, p_2) = (13, 11, 61)$ and $(37, 67, 97)$, the corresponding check program reconstructs every relevant principal ideal, asks `bnfisprincipal` for a generator α , verifies that (α) is the reconstructed ideal, and then tests $\alpha \cdot \text{conj}(\alpha) = 2^r$ in exact integral-basis coordinates. The verification of a returned generator is exact; the claim that r is the first PIP endpoint remains conditional on GRH.

7. Reproducibility checklist

- Confirm that the primes and congruence conditions pass the program's input checks.
- Record the PARI/GP version and the values of p_1 , p_2 , e_1 , e_2 , `start_n`, and `max_n`.
- For large-field searches, retain the printed class-group invariants, first endpoint, full exponent vectors, principality verification, and support-test result.
- For quartic benchmark runs, retain the line `bnfcertify(Tbnf) = 1`.
- For the two exceptional triples, retain the printed integral-basis coordinates of each successful generator and the exact-norm success message.
- Keep the conditional and unconditional conclusions separate in any report of the computation.

8. PARI/GP documentation

The official PARI/GP reference documents `bnfinit`, `bnfcertify`, and `bnfisprincipal` in the general-number-fields catalogue: pari.math.u-bordeaux.fr/dochtm/html/General_number_fields.html.